0580_s23_qp_41_Q8

8 A tailor makes x dresses and y shirts in one week.
In one week

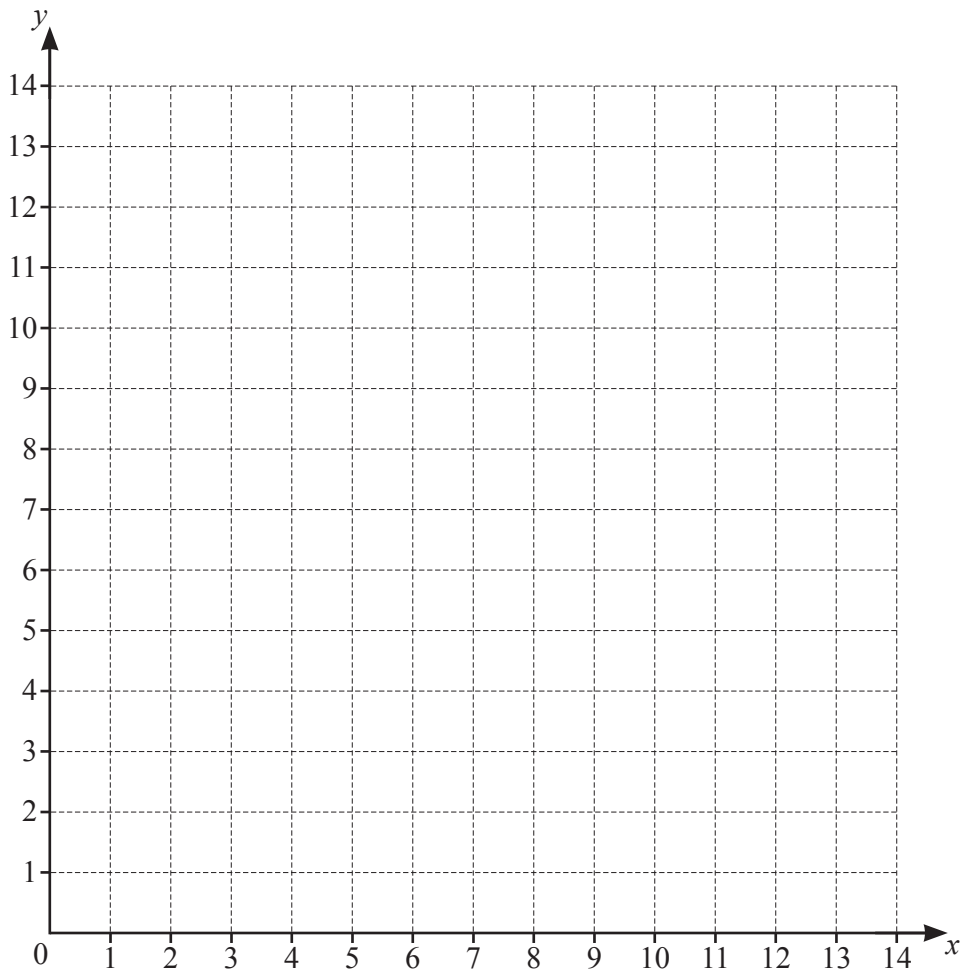
- he makes at least 4 dresses
- he makes no more than 7 shirts
- he makes less than 14 dresses and shirts altogether
- the number of shirts he makes is more than $\frac{2}{3}$ of the number of dresses.

One of the inequalities that shows this information is $x \geqslant 4$.

(a) Write down the other three inequalities in x and/or y.

.... [3]

(b)



On the grid, draw 4 straight lines and shade the unwanted regions to show these inequalities.
Label the region R that satisfies the 4 inequalities. [6]

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(c) Use your diagram to find the smallest number of dresses and the smallest number of shirts the tailor makes in one week.

..... dresses and ..... shirts [1]

(d) The profit the tailor makes on one dress is \$10 and the profit on one shirt is \$6.

Use your diagram to find the largest profit the tailor can make in one week.

\$ ..... [2]